Titration Calculations

Exam code: 0620 & 0971

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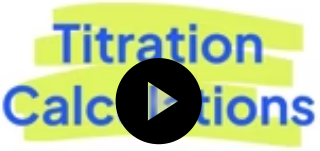
Updated on 25 September 2024



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Titration calculations

Extended tier only

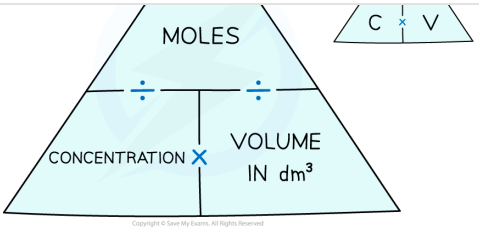
- Titrations are a method of analysing the **concentration** of solutions
  - Acid-base titrations are one of the most important kinds of titrations
- Titrations can determine exactly how much alkali is needed to neutralise a quantity of acid – and vice versa
- You may be asked to calculate:
  - The **moles** present in a given amount
  - The **concentration** or **volume** required to **neutralise** an acid or a base
- Once a titration is completed and the average titre has been calculated, you can calculate the unknown variable using the formula triangle as shown below

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Formula triangle showing the relationship between concentration, number of moles and volume of liquid

Worked Example

A solution of 25.0 cm^3 of hydrochloric acid was titrated against a solution of 0.100 mol/dm^3 NaOH.

12.1 cm^3 of NaOH was required for a complete reaction.

Determine the concentration of the acid.

Answer:

- **Step 1:** Write the equation for the reaction:
  - **HCl (aq) + NaOH (aq) → NaCl (aq) + H_2O (l)**
- **Step 2:** Calculate the number of moles of the NaOH
  - Moles = $(\frac{\text{volume}}{1000})$ x concentration
  - Moles of NaOH = 0.012 dm^3 x 0.100 mol / dm^3 = 1.21 x 10^{-3} mol
- **Step 3:** Deduce the number of moles of the acid
  - Since the acid reacts in a 1:1 ratio with the alkali, the number of moles of HCl is also 1.21 x 10^{-3} mol
  - This is present in 25.0 cm^3 of the solution (25.0 cm^3 = 0.025 dm^3)
- **Step 4:** Find the concentration of the acid
  - Concentration = $\frac{\text{moles}}{\text{volume } (dm^3)}$
  - Concentration of HCl = $\frac{1.21 \times 10^{-3} \text{ mol}}{0.025 \text{ dm}^3}$ = **0.0484 mol/dm^3**

Worked Example

25.00 cm^3 of 0.15 mol / dm^3 barium hydroxide, $Ba(OH)_2$, was required to neutralise 12.80 cm^3 of nitric acid, HNO_3 , during a titration. Calculate the concentration of HNO_3 that was used. Give your answer to 2 decimal places.

$$Ba(OH)_2 \text{ (aq)} + 2HNO_3 \text{ (aq)} \rightarrow Ba(NO_3)_2 \text{ (aq)} + 2H_2O \text{ (l)}$$

Answer:

- **Step 1:** Calculate the number of moles of barium hydroxide

moles of nitric acid
  - Moles of nitric acid = $3.75 \times 10^{-3} \times 2 = 7.5 \times 10^{-3}$
  - The number of moles must be multiplied by 2 due to the 1:2 ratio
- **Step 3:** Calculate the concentration of nitric acid
  - Concentration of nitric acid = $\frac{7.5 \times 10^{-3}}{0.0128}$ = **0.59 mol / dm^3 to 2 dp**
- Remember to convert cm^3 to dm^3 by dividing by 1000

Worked Example

Calculating volume

Calculate the volume of 0.50 mol / dm^3 nitric acid, HNO_3, required to neutralise 25.00 cm^3 of 0.80 mol / dm^3 potassium hydroxide, KOH. Give your answer in cm^3.

$$KOH\ (aq) + HNO_3\ (aq) \rightarrow KNO_3\ (aq) + H_2O\ (l)$$

Answer:

- **Step 1:** Calculate the number of moles of potassium hydroxide
  - Moles of potassium hydroxide = concentration x volume (dm^3) = 0.80 x 0.025 = 0.02 mols
- **Step 2:** Using the equation, calculate the number of moles of nitric acid
  - Moles of nitric acid = 0.02 mols
  - The ratio is 1:1 so the number of moles of nitric acid is the same
- **Step 3:** Calculate the volume of nitric acid in cm^3
  - Volume of nitric acid = $\frac{\text{moles}}{\text{concentration}} = \frac{0.02}{0.50}$ = 0.040 dm^3
  - Volume of nitric acid = 0.040 dm^3 x 1000 = 40 cm^3

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